

Development 140, 3743–3753 (2013) doi:10.1242/dev.095810
 © 2013. Published by The Company of Biologists Ltd

Dual role of delta-like 1 homolog (DLK1) in skeletal muscle development and adult muscle regeneration

Ditte Caroline Andersen^{1,2}, Jorge Laborda³, Victoriano Baladron³, Moustapha Kassem^{4,5}, Søren Paludan Sheikh^{1,6} and Charlotte Harken Jensen^{1,*}

SUMMARY

Muscle development and regeneration is tightly orchestrated by a specific set of myogenic transcription factors. However, factors that regulate these essential myogenic inducers remain poorly described. Here, we show that delta-like 1 homolog (*Dlk1*), an imprinted gene best known for its ability to inhibit adipogenesis, is a crucial regulator of the myogenic program in skeletal muscle. *Dlk1*^{−/−} mice were developmentally retarded in their muscle mass and function owing to inhibition of the myogenic program during embryogenesis. Surprisingly however, *Dlk1* depletion improves *in vitro* and *in vivo* adult skeletal muscle regeneration by substantial enhancement of the myogenic program and muscle function, possibly by means of an increased number of available myogenic precursor cells. By contrast, *Dlk1* fails to alter the adipogenic commitment of muscle-derived progenitors *in vitro*, as well as intramuscular fat deposition during *in vivo* regeneration. Collectively, our results suggest a novel and surprising dual biological function of DLK1 as an enhancer of muscle development, but as an inhibitor of adult muscle regeneration.

KEY WORDS: Delta-like 1 homolog, Muscle development, Muscle regeneration, Muscle stem cells, Mouse

INTRODUCTION

Insight into factors that promote or inhibit normal muscle regeneration is crucial to open up new possibilities for treating muscle diseases, such as lethal muscular dystrophies and age-dependent sarcopenia, that cause an enormous economic burden in the increasingly aged population (Saini and Stewart, 2006). Normal muscle develops and regenerates completely from satellite cells (referred to as ‘classical’ muscle stem cells), but the restorative capacity is highly reduced or eradicated in both diseased and atrophied muscles (Evans, 2010). Both skeletal muscle development and regeneration is known to embrace a cascade of temporarily expressed specific transcriptional regulators, such as *Pax7*, *MyoD*, myogenin, *Mef2c*, *Myf5*, etc., in satellite cells (Zammit et al., 2006). Thus, although recapitulation of the fetal myogenic program *per se* enables restoration of the damaged adult muscle, this mechanism seems to be inadequate in muscle disease and in aged individuals (Evans, 2010; Saini and Stewart, 2006; Zammit et al., 2006).

We have previously identified delta-like 1 homolog (*Dlk1*) as a gene highly induced during normal muscle development and regeneration in both human and mouse (Andersen et al., 2009), yet its biological function in these processes remains unknown. *Dlk1* is a paternally expressed imprinted gene that encodes for membrane and secreted protein variants, which are members of the NOTCH receptor and ligand epidermal growth factor (EGF)-like protein family (Laborda et al., 1993). Alternative splicing generates various

forms of membrane-spanning DLK1 proteins (Smas et al., 1994) with some variants encompassing a protease recognition site that enables processing and shedding (Jensen et al., 1994; Mei et al., 2002). Although DLK1 has been suggested to interact and function through different surface proteins, including NOTCH1, this is still controversial and not well understood. During development, *Dlk1* is expressed in developing myofibers and associated satellite cells, whereas its expression is almost completely abolished in adult muscle in both human and mouse (Andersen et al., 2009; Floridon et al., 2000). Likewise, diseased and regenerating muscle express high levels of *Dlk1*, though the number of DLK1⁺ satellite cells in these scenarios differ greatly, with only very few appearing during muscle regeneration of healthy muscle (Andersen et al., 2009). By contrast, numerous DLK1⁺ non-myogenic cells emerge in the interstitium between the myofibers upon muscle damage in human and mouse (Andersen et al., 2009; Waddell et al., 2010). Thus, as DLK1 can act in a cell autonomous or non-cell autonomous manner, it may affect myogenic precursor cells regardless of which type of cell expresses DLK1 in the muscle. However, the biological function of the muscle-residing DLK1⁺ cell populations still remains incompletely understood. In this study, we performed a detailed comparison of the skeletal muscle phenotypes of wild-type and *Dlk1* knockout mice during embryogenesis and adulthood, as well as during adult muscle regeneration, and thereby unraveled a highly unusual dual role of *Dlk1* in skeletal muscle development and regeneration. Our data suggests that inhibitors of *Dlk1* function may be useful to facilitate adult muscle regeneration.

MATERIALS AND METHODS

Animals

Dlk1^{−/−} and *Dlk1*^{+/+} colonies were maintained by homozygous breeding (Raghuveer et al., 2008) and backcrossed to C57BL/6 mice every third generation in order to ensure maximal genetic diversity and avoid accumulation of spontaneous mutations. Tail or ear DNA was isolated using a DNeasy Kit (Qiagen) and genotype analysis was performed by PCR amplification using the following primers: 5′*Dlk1*_F, CCAAATTGTCTATAGTCTCCCTC; 5′*Dlk1*_R, CTGTATGAAGAGGACCAAGG; 5′Neo_F, TTGAACAAGATGGATTGCACGAGG; 5′Neo_R, GGCT-

¹Laboratory of Molecular and Cellular Cardiology, Department of Clinical Biochemistry and Pharmacology, Odense University Hospital, Winsloewparken 21 3rd, 5000 Odense C, Denmark. ²Institute of Clinical Research, University of Southern Denmark, Odense, Denmark. ³Department of Inorganic and Organic Chemistry and Biochemistry, Medical School, Regional Center for Biomedical Research, University of Castilla-La Mancha, Avenida de Almansa 14, 02006 Albacete, Spain. ⁴Department of Endocrinology and Metabolism, Odense University Hospital, Odense, Denmark. ⁵Stem Cell Unit, Department of Anatomy, College of Medicine, King Saud University, Riyadh 11461, Saudi Arabia. ⁶Department of Cardiovascular and Renal Research, Institute of Molecular Medicine, University of Southern Denmark, Odense, Denmark.

* Author for correspondence (charken@health.sdu.dk)

GGCGCGAGCCCCTGATGCTCT, and Taq DNA polymerase (Invitrogen). Animals were housed in plastic cages with a 12/12 hour light/dark cycle, and fed *ad libitum* with a normal chow (10% fat, 20% protein, 70% carbohydrate). Specific characteristics of the animals have been described previously (Raghunandan et al., 2008) and are in agreement with others (Moon et al., 2002). For all animal experiments we used age- and gender-matched animals, as indicated. All procedures were approved by the Danish Council for Supervision with Experimental Animals (#2010/561-1792).

Skeletal muscle regeneration model

Skeletal muscle lesion was performed as previously described (Andersen et al., 2009) in 12-week-old female and male *Dlk1*^{-/-} and *Dlk1*^{+/-} C57BL/6 mice. Briefly, hindlimb muscles of anesthetized mice were carefully exposed and two wounds were introduced starting from the outer fascia to the widest part of musculus (m.) gastrocnemius until reaching the calfbone, then the skin incision was sutured. The m. gastrocnemius was dissected out from mice 0–14 days following muscle damage. For immunohistochemistry, the tissues were snap frozen. For qRT-PCR, the m. gastrocnemius was carefully dissected with removal of fat and tendons, and for each mouse the two m. gastrocnemii were pooled and kept in RNA Lysis Buffer (Applied Biosystems, ABI, Foster City, CA, USA) until further analysis.

Dissection of limb buds

Hindlimb buds were dissected using stereomicroscopy from embryonic day (E)11.5 *Dlk1*^{-/-} and *Dlk1*^{+/-} C57BL/6 embryos and all buds from one litter ($n^*=7$ –10) were pooled and used as $n=1$.

Isolation and culture of muscle-derived cells

Muscle-derived cells (MDCs) were retrieved (each experiment comprising two m. gastrocnemii from each of two mice) using a previously described protocol (Andersen et al., 2009; Andersen et al., 2008). Briefly, the gastrocnemius muscles were dissected and visible fat, as well as large vessels, were removed. Tissues were subsequently rinsed with Hank's Balanced Salt Solution (HBSS) + 100 IU/ml penicillin-streptomycin (1% PS) and minced extensively with scalpels before digestion with 0.30% Type II collagenase (Worthington, UK) at 37°C for 40 minutes. Cells were released by gentle trituration DMEM + 20% fetal calf serum (FCS) + 1% PS (all GIBCO products, Life Technologies, Denmark) at 37°C, and a single cell suspension was obtained by serially filtering the cell samples through 100- and 40- μ m cell strainers. Isolated cells were pelleted by centrifugation, and the cells were plated on extracellular matrix (ECM; E1270 Sigma-Aldrich Europe)-coated dishes at very low cell density. After 48 hours of culture, 50% confluent cells were passaged, counted by Coulter counting, and plated at 3000 cells/cm². For myogenic differentiation experiments, cells were cultured in DMEM + 20% FCS + 1% PS for 72 hours (used as proliferative samples) and myogenic differentiation was then allowed to occur in DMEM + 2% horse serum + 1% PS for an additional period of 72 hours (used as differentiated samples in experiments). For adipogenic differentiation, cells were cultured as recently described (Andersen et al., 2013). Briefly, adipogenic differentiation was induced in confluent MDCs by the addition of DMEM containing 1% PS, 5% FCS, 1 mM dexamethasone, 0.5 mM 3-isobutyl-1-methylxanthine, 10 mg/ml insulin and 5 μ M roglitazone for 3 days and further maintained for 7 days in DMEM containing 1% PS, 5% FCS and 10 mg/ml insulin.

Overexpression of DLK1 in the myogenic cell line C2C12

The myogenic cell line C2C12 (ATCC) was kept in agreement with general guidelines and maintained in DMEM + 10% FCS + 1% PS at 37°C in humidified 5% CO₂. For all experiments, C2C12 cells were seeded at 10,000 cells/cm² and cultured for 48 hours before further treatment. For transfection experiments, cells were equilibrated in serum-free medium for 1 hour prior to transfection for 4 hours with 1 μ g/12-well pCD2-Dlk1FL (containing full-length mouse *Dlk1* cDNA (Baladrón et al., 2002) or control pCD2 vector using the Lipofectamine 2000 protocol (Life Technologies, Denmark). Immediately following transfection, cells were induced to differentiate in DMEM + 2% horse serum + 1% PS for 5 days. Conditioned medium from pCD2-Dlk1FL or control pCD2 vector transfected cells was added to C2C12 cells in order to induce differentiation in the presence/absence of soluble mouse DLK1 for 5 days. The dose-response of

C2C12 cells to DLK1 during differentiation was analyzed by addition of purified DLK1 (Jensen et al., 1994) at a concentration of 0, 2.5, 5 or 10 μ g/ml for 5 days. Myofiber diameter was measured using Adobe Photoshop on pictures obtained from myosin-stained cultures ($n=4$).

Histology, immunofluorescence and immunohistochemistry

Immunocytochemistry was performed as previously described (Andersen et al., 2009; Andersen et al., 2008). Cultured cells were gently rinsed in TBS, fixed in 4% neutral buffered formaldehyde (NBF) (10 minutes), permeabilized for 10 minutes in 0.3% Triton X-100 in TBS, blocked in 2% bovine serum albumin (BSA) in TBS (10 minutes), and incubated for 2 hours with primary antibodies (see below) diluted in 1% BSA in TBS. Secondary antibodies used were Alexa 555- or 488-conjugated donkey anti-IgG (1:200, Molecular Probes), and mounting medium contained DAPI (Vectashield, Vector Labs).

For histology and IHC, muscles were snap-frozen in Tissue Tek or fixed overnight in 4% NBF and embedded in paraffin. Paraffin-embedded sections were deparaffinized and rehydrated before antigen demasking for IHC (heating in Tris-EGTA; pH 9). Blocking was performed (10 minutes) in 2% BSA in TBS, and primary antibodies diluted in 1% BSA in TBS were applied overnight at 4°C. Secondary antibodies used were Alexa 555-, 488- or 647-conjugated donkey anti-IgG (1:200, Molecular Probes), and mounting medium contained DAPI (Vectashield, Vector Labs). For morphological analyses, sections from fixed or snap-frozen gastrocnemius muscle were stained with HE or Sirius Red.

Microscopic examinations were performed using a Leica DMI4000B Cool Fluo Package instrument equipped with a Leica DFC340 FX Digital Camera. In all experiments, exposure (camera settings) and picture processing (brief adjustment of contrast/brightness and color balance by using Photoshop) were applied equally to sample sections and controls (isotype or no primary antibody present). The number of fibers were counted on pictures showing cross-sections of paraffin-embedded muscles at day 14 following lesion ($n=5$ –7). Sections were taken from the site where m. gastrocnemius is broadest. Antibodies used were: mouse anti-myosin (1:200; Developmental Studies Hybridoma Bank), rabbit anti-mouse DLK1 (1:2000; in-house) (Bachmann et al., 1996), rat anti-CD45 (1:50, BD Pharmingen), rabbit anti-collagen I (1:100, Abcam, ab34710); goat anti-desmin (1:50, Santa Cruz Biotechnology), rat anti-laminin-2 (1:50, Abcam), mouse anti-actinin (1:200, Sigma-Aldrich), mouse anti-slow twitch myosin (1:200, Sigma-Aldrich) and mouse anti-myogenin (1:100, DAKO).

Rotarod

Twelve-week-old female *Dlk1*^{-/-} ($n=7$; one animal died before testing) and *Dlk1*^{+/-} ($n=8$) mice were pre-trained on an LE8200 Rotarod system (Panlab, Harvard Apparatus) at days -8 and -6 prior to muscle lesion. Pre-training comprising three sessions during which all animals were successfully trained to stay and run on the rod for one minute at a fixed speed of 24 rpm. Trials were performed during muscle regeneration at day -1, +3, +6 and +14 using a constant speed of 24 rpm and a maximum testing time of 5 minutes. Each trial was performed three times and animals rested for one hour in between. The latency to fall (seconds) from the rod was recorded automatically and all three scores per mouse per day were averaged and recorded as latency to fall (in seconds) for each mouse. Rotarod performance was also tested in adult *Dlk1*^{-/-} and *Dlk1*^{+/-} mice to examine muscle phenotype using the rotarod setup described above.

Oil Red O staining and quantification

Oil Red O staining was performed as recently described (Andersen et al., 2013). Differentiated cells were rinsed in PBS, fixed (30 minutes) in 10% NBF (VWR, Herlev, Denmark), carefully washed three times in TBS, and stored at 4°C until analysis. TBS was discarded, and cells were briefly exposed to 3% 2-propanol (1 second) and completely air-dried before incubation (1 hour) with filtered Oil Red O solution (Sigma-Aldrich, Europe) prepared according to the manufacturer's recommendation. Stained cells were rinsed gently with water to remove any precipitates, and examined microscopically with a Leica DMI4000B Cool Fluo Package instrument (Leica Microsystems, Ballerup, Denmark) equipped with a Leica DFC340 FX Digital Camera. In all experiments, camera settings and picture processing were applied equally to all sample sections. Oil Red O

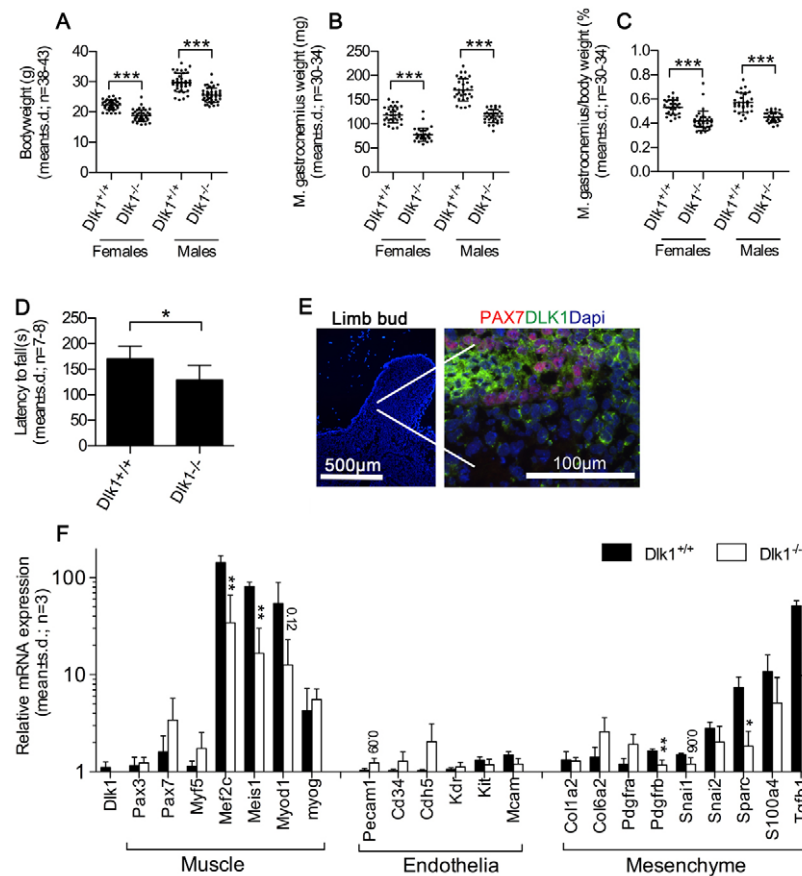


Fig. 1. *Dlk1* is required for normal skeletal muscle mass and function. (A–C) Irrespective of gender, adult (12-week-old) *Dlk1* null mice displayed a 14–16% decrease in body weight (A) and 32–35% less m. gastrocnemius mass (B) resulting in a 21% reduction in m. gastrocnemius-to-body mass ratios (C) compared with wild-type controls. (D) The rotarod test was used to assess muscle function performance in adult *Dlk1*^{+/+} and *Dlk1*^{-/-} mice. The latency to fall (seconds) off the rotating rotarod is expressed as the mean ± s.d. of three trials for each mouse within the two groups. (E) Double immunofluorescence of DLK1 and PAX7 in hindlimb buds from a *Dlk1*^{+/+} E10.5 embryo showing colocalization in cells within the embryonic muscle anlage of normal wild-type mice. (F) qRT-PCR of E11.5 hindlimb buds from *Dlk1*^{-/-} and *Dlk1*^{+/+} embryos (n=3, each experiment consisting of a pool of seven to ten embryos) showing differences in relative mRNA expression levels of various muscle-, vascular- and mesenchymal-related genes. qRT-PCR raw data were normalized against *Gapdh* and *Gusb*, which were stably expressed (qBase-M: 0.469; CV 0.163). Error bars indicate s.d. Statistical significance (tested by Student's *t*-test) is indicated (**P*<0.05, ***P*<0.01, ****P*<0.0001).

quantification was performed as follows: stained cells were allowed to dry, and Oil Red O was eluted by incubation (10 minutes) with 100% 2-propanol. Eluted staining solution was then transferred to a 96-well microtiter plate and the optical density at 510 nm was read in triplicate using an ELISA plate reader. Empty wells were withdrawn during the entire fixing, staining and quantification procedure, and used as blank controls. Oil Red O staining was performed in four independent biological experiments, each comprising triplicate wells on 12-well plates. Each of these wells was stained with Oil Red O and measured in technical triplicates. The overall mean was used for analysis. For Oil Red O staining of cryo-embedded tissue, sections were air-dried, fixed in 4% NBF, and otherwise processed as described above for cells.

Coulter cell counting

Cultured cells were gently detached with 0.25% trypsin-EDTA (Gibco), pelleted and re-suspended in Hank's Balanced Salt Solution (HBSS, Lonza) + 10% FCS + 1% PS. Cell number and cell diameter were determined using a Beckman Coulter Counter Z2 (Ramcon) fitted with a 100 µm aperture. The size range of particles counted was set at 11–27 µm, and counting was performed in the indicated number of independent experiments each comprising triplicate measurements.

Relative quantitative real-time PCR (qRT-PCR)

Total RNA was extracted from cell cultures or muscle tissue using the semi-automated 6100 Nucleic Acid Prep Station system, according to manufacturer's instructions (Applied Biosystems). For cDNA synthesis, 0.3–0.4 µg of total RNA was reverse transcribed with a High Capacity cDNA RT Kit (#4368813, Applied Biosystems), and qRT-PCR reactions were performed in technical triplicates using commercially available Taqman assays. The PCR was run on a 7900HT Fast Real-time PCR system (Applied Biosystems) and robust and valid qRT-PCR data were obtained by normalizing the raw data against multiple stably expressed endogenous control genes as determined by the qBase Plus platform (Hellemans et al., 2007; Vandesompele et al., 2002).

Statistical analyses

All analyses comprised at least three independent experiments. One- or two-way ANOVA (with or without repeated measure), and two-tailed *t*-tests were performed as indicated using GraphPad Prism to test significance (*P*<0.05).

RESULTS

Dlk1-null mice are developmentally impaired in their muscle mass and function owing to inhibition of the myogenic program

Our previous results have demonstrated DLK1 expression in skeletal muscle during development, regeneration and disease (Andersen et al., 2009; Floridon et al., 2000). However, the functional significance of *Dlk1* in skeletal muscle is not well understood. Herein, we therefore performed a detailed assessment of the muscle phenotype in *Dlk1*^{+/+} and *Dlk1*^{-/-} mice. As previously reported by others (Moon et al., 2002), adult *Dlk1*-null mice were clearly growth retarded compared with gender- and age-matched wild-type controls. The body weights of adult *Dlk1*^{-/-} female and male mice were significantly (*P*<0.0001) reduced to a similar extent (15.8% and 14.0%, respectively; mean; n=38–43) (Fig. 1A). Surprisingly, we found an even greater (34.8% and 31.9%; mean; *P*<0.0001; n=30–34) decrease in the m. gastrocnemius mass in *Dlk1*^{-/-} females and males, respectively, compared with their corresponding wild types (Fig. 1B). Thus, the m. gastrocnemius-to-body mass ratios were dramatically reduced (*P*<0.0001) in female (average 21.3%; n=30–34) and male (average 21.2%; n=30) *Dlk1*-null versus wild-type mice (Fig. 1C), suggesting that lack of *Dlk1* causes a dramatic inhibitory effect on muscle mass in adult mice. This diminished muscle mass was mirrored by a poor functional

performance on the rotarod apparatus; *Dlk1*^{+/-} mice performed significantly better (24%) than *Dlk1*-null mice (Fig. 1D). As a higher body mass usually correlates with a shorter active rotation time (McFadyen et al., 2003), this figure might be an underestimation. No apparent difference was observed in muscle morphology between *Dlk1*^{-/-} and *Dlk1*^{+/-} mice, and no centrally located nuclei were observed, suggesting normal maturation of the myofibers (data not shown). It is known that *Dlk1* is absent in normal adult skeletal muscle (Andersen et al., 2009). Thus, it seems likely that *Dlk1* does not exert any effect at this adult stage, at least in a cell-autonomous manner, considering that soluble DLK1 protein variants produced and secreted by other cells may be present. Even so, we hypothesized that the presence of high *Dlk1* levels described in embryonic mouse skeletal muscle (Andersen et al., 2009; Floridon et al., 2000; Yevtodiynenko and Schmidt, 2006) could account for the functional muscle phenotype we observed in *Dlk1*^{-/-} adult muscle (Fig. 1A-E). To evaluate this, we first identified DLK1 expression in embryonic muscle anlage at the protein level using DLK1-PAX7 double immunofluorescence staining (Fig. 1E). We next performed a detailed molecular analysis of the myogenic gene expression program in dissected hindlimb buds from E11.5 *Dlk1*^{+/-} and *Dlk1*^{-/-} mice (Fig. 1F). Whereas *Pax3*, *Pax7* and *Myf5* levels were slightly higher in *Dlk1*^{-/-} mice compared with wild type, we found a clear and significant suppression of the *Mef2c*, *Meis1* and *Myod1* genes (Fig. 1F). These three latter transcription factors are known to cooperate in order to activate the myogenic transcriptional program (Black and Olson, 1998). Our data thus support the conclusion that the observed muscle-retarded phenotype in *Dlk1*^{-/-} mice is likely to be due to a restrained myogenic differentiation program in muscle progenitors during skeletal muscle development. Notably, we also observed a lower expression in *Dlk1*-null limb buds of genes associated with mesenchyme, whereas a tendency for higher amounts of some vascular transcripts was seen (Fig. 1F), which is in agreement with the known involvement of *Dlk1* in bone and vascular morphogenesis (Abdallah et al., 2011; Chen et al., 2011; Rodríguez et al., 2012).

***Dlk1*-depleted heterogeneous muscle-derived cell cultures exhibit an enhanced myogenic phenotype**

Although *Dlk1* is downregulated after birth and its expression is almost abolished in adult skeletal muscle, it is re-expressed during muscle regeneration (Fig. 2A) (Andersen et al., 2009). We have shown that *Dlk1* expression peaks at day 7 following a knife-cut lesion in m. gastrocnemius (Andersen et al., 2009), yet the role of this high *Dlk1* induction during muscle regeneration remains to be determined. In agreement with our previous results and those of others (Waddell et al., 2010), we found that DLK1 localizes mainly to interstitial cells (outside laminin⁺ myofibers) (Fig. 2A) of a non-myogenic cell lineage (Andersen et al., 2009) during muscle regeneration. To elucidate biological effects of *Dlk1* in muscle regeneration, we established heterogeneous cultures of proliferating and differentiating muscle-derived mononuclear cells (MDCs; containing muscle progenitors and interstitial cells), isolated from day 7 post-lesion *Dlk1*^{-/-} and *Dlk1*^{+/-} m. gastrocnemius (Fig. 2B). Flow cytometry of proliferating *Dlk1*^{+/-} MDC cultures revealed that 16.6±2.1% (mean±s.d.; *n*=3) of the MDCs expressed DLK1 (Fig. 2C) but, as shown by a 52% reduction in *Dlk1* mRNA levels and supported at the protein level by immunocytochemistry, *Dlk1* expression decreased with *in vitro* myogenic differentiation, as expected (Fig. 2B,D). Neither *Dlk1* mRNA nor protein was detected

in *Dlk1*^{-/-} muscle tissue and MDC cultures (Fig. 2A-D), thus verifying our experimental setup. As seen during muscle embryogenesis, we found several important myogenic transcription factors, as well as structural proteins, to be regulated by *Dlk1* (Fig. 2E). However, most unexpectedly, the expression pattern was inverse to that observed during muscle development (Fig. 1). Thus, *Dlk1*^{-/-} MDCs, compared with *Dlk1*^{+/-} MDCs, expressed significantly higher levels of the myogenic transcription factors *Pax3*, *Pax7* and *Myf5*, and showed a tendency to increased expression of *Mef2c*, *Myod1* and *Myog* (Fig. 2E). A similar pattern of enhanced myogenesis in *Dlk1*^{-/-} MDC cultures was seen for the major structural muscle proteins *Myh7*, *Tnnt2* and *Des*. By contrast, *Ncam1* and *Cd34*, two genes also often associated with myogenic precursors (Zammit et al., 2006), but also with other cells, seemed not to be affected by *Dlk1* (Fig. 2E). These results thus suggested that *Dlk1* exerts an unexpected inhibitory role on the myogenic differentiation program of adult muscle progenitors *in vitro*. We also noticed an increased level of proliferating cell nuclear antigen (*Pcna*) in *Dlk1*-depleted MDCs (Fig. 2F), which is in agreement with a recent study showing growth inhibition of DLK1-overexpressing myogenic cells (Waddell et al., 2010). Visualization of myofibers (Fig. 2G) in the *Dlk1*^{-/-} and *Dlk1*^{+/-} MDCs supported this conclusion, as *Dlk1*^{-/-} myofibers appeared more mature and larger than those present in *Dlk1*^{+/-} cultures; this was quantitatively confirmed by an increased expression level of the structural muscle protein myosin (Fig. 2H). These results thus indicate that *Dlk1*, at least *in vitro*, plays a regulatory role during adult muscle remodeling in which depletion of *Dlk1*, unexpectedly, promotes a myogenic program.

Additionally, we found a clear and significant downregulation of vascular genes (supplementary material Fig. S1B) as well as a modest increase in some mesenchymal genes, whereas others remained unchanged (supplementary material Fig. S1C). *Dlk1* may thus play a dual regulatory role, increasing vascular but decreasing myogenic commitment during adult muscle regeneration, whereas mesenchymal remodeling seems to be unaffected.

***In vivo* muscle regeneration is improved in *Dlk1*-deficient animals**

To elucidate whether our findings demonstrate a biological mechanism during *in vivo* adult muscle regeneration, we performed a detailed series of knife-cut lesions in *Dlk1*^{-/-} and *Dlk1*^{+/-} gastrocnemius muscles. Because no difference was observed in the muscle phenotype between genders (Fig. 1), we designed our experimental setup to comprise high biological diversity with two females and two males (*n*=4) of each strain and analyzed muscle regeneration during 14 days.

Similarly to previous observations (Andersen et al., 2009), we observed that *Dlk1* mRNA increased in *Dlk1*^{+/-} gastrocnemius at day 4 post-lesion, peaked at day 7 and, except for a small peak at day 10, it thereafter declined to reach basal levels at day 14 (Fig. 3A). To evaluate the overall regeneration process in *Dlk1*^{-/-} and *Dlk1*^{+/-} lesioned muscle, we examined Hematoxylin and Eosin (HE)-, collagen/CD45-, DLK1/myosin- and Sirius Red-stained sections and performed various qRT-PCRs. In both strains, numerous degrading myofibers (supplementary material Fig. S2A), as well as an influx of inflammatory CD45⁺ cells (Fig. 3B), could be seen at day 2 post-lesion, reflecting an initial phase of muscle degeneration and inflammation, a result that was confirmed quantitatively by qRT-PCR for caspase 3 (pre-apoptotic marker) and *Cd45* (*Ptprc*), *Thyl1*, *Cd14* and *Cd68* (hematopoietic markers) (Fig. 3C). In parallel, a significant increase was found for

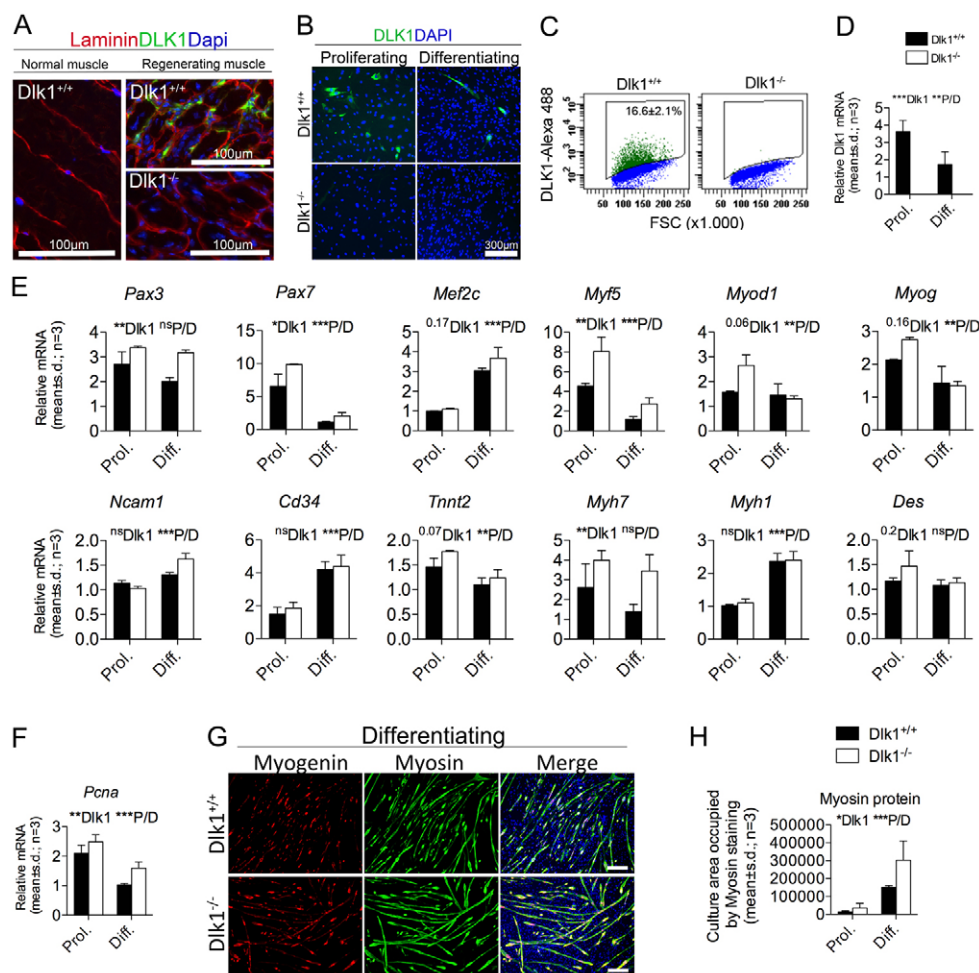
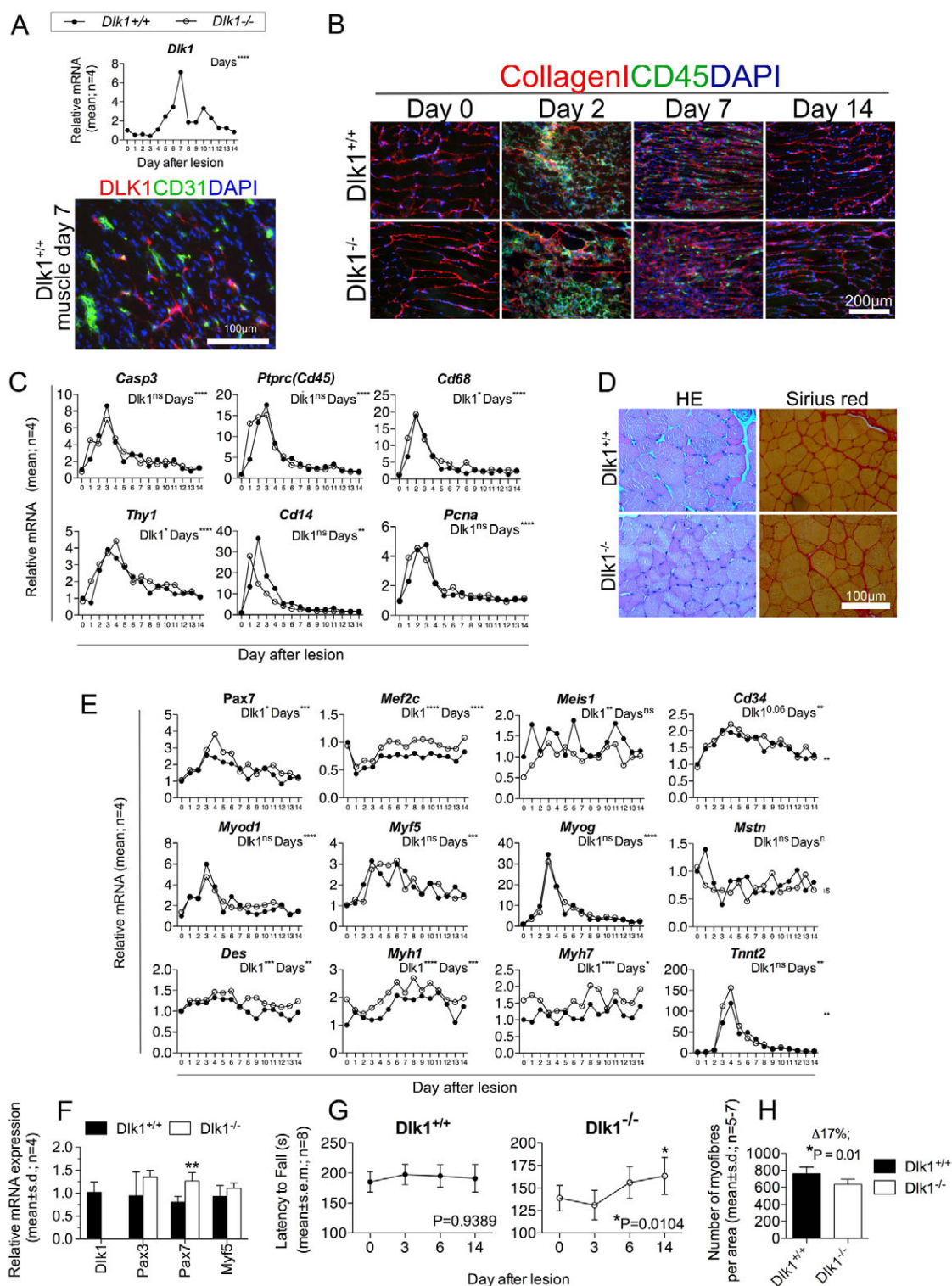


Fig. 2. *In vitro* myogenesis is enhanced in *Dlk1*-depleted muscle-derived cultures. (A) Immunohistological analysis of DLK1 and laminin in normal and regenerating (at day 7 post-lesion) muscle. DLK1 staining is observed mainly in non-myogenic interstitial cells. (B–F) Proliferating (P) and differentiating (D) muscle-derived mononuclear cells (MDCs) isolated from m. gastrocnemius at day 7 post-lesion was analyzed by various assays: (B) Immunocytochemistry of cultured MDCs showed a decrease in DLK1⁺ cell numbers during *in vitro* differentiation of wild-type cells, whereas no DLK1⁺ cells were observed in *Dlk1*^{-/-} MDCs. (C) Flow cytometry for DLK1 in proliferating MDCs. Plots represent one of three independent experiments and the percentage of DLK1⁺ cells is given as the mean±s.d. Green, DLK1⁺ cells; blue, DLK1⁻ cells. (D) Relative qRT-PCR levels of *Dlk1*. (E) Normalized mRNA levels of muscle-related genes in the proliferating and differentiated MDCs were determined by qRT-PCR. Most myogenic mRNAs were expressed at a higher level in *Dlk1*^{-/-} versus *Dlk1*^{+/+} cultures. (F) Normalized relative mRNA levels of proliferating cell nuclear antigen (*PcnA*) in MDC cultures. For D–F, data were normalized against *18S* and *Tbp* (qBase-M: 0.258; CV 0.089). (G,H) Immunocytochemistry of differentiated MDCs showing that *Dlk1*^{-/-} fibers appear larger than *Dlk1*^{+/+} fibers and express more myosin overall (expressed as culture area stained by myosin and quantified using CellProfiler). Scale bars: 300 µm. For all graphs, error bars indicate s.d.; statistical significance was tested by a two-way ANOVA (genotype: *Dlk1*; proliferation/differentiation (P/D); **P*<0.05, ***P*<0.01, ****P*<0.0001, ns, not significant).

proliferating cell nuclear antigen (*PcnA*) (Fig. 3C), suggesting activation of muscle-residing cells, which is in agreement with observations by others (Yan et al., 2003). At day 7 following lesion, numerous interstitial DLK1⁺/CD31⁻ (Fig. 3A), DLK1⁺/Myosin⁻ and DLK1⁺/actinin⁻/laminin⁻ cells (supplementary material Fig. S2B) were observed exclusively in the regenerating zone of *Dlk1*^{+/+} muscle, though many interstitial cells were seen in *Dlk1*^{-/-} sections as well (Fig. 2A; supplementary material Fig. S2B). The regenerative phase was confirmed by the presence of many newly formed actinin⁺ and myosin⁺ myofibers with centrally localized nuclei (supplementary material Fig. S2B). At day 14, the overall muscle architecture was re-established in *Dlk1*^{-/-} and *Dlk1*^{+/+} muscles (Fig. 3A,D; supplementary material Fig. S2A). Total collagen deposition as shown by Sirius Red staining at days 0–14

was equally distributed in samples from both strains (Fig. 3D; supplementary material Fig. S2C), and except for a modest increase (days 2–4) in collagen I mRNA levels in *Dlk1*-null muscles, testing of 13 genes associated with mesenchyme and fibrosis did not reveal any difference between *Dlk1*^{-/-} and *Dlk1*^{+/+} animals (data not shown). These findings thus revealed that *Dlk1*^{-/-} animals fully accomplish skeletal muscle regeneration with no overall change in inflammatory response and fibrotic scarring.

To investigate quantitatively whether *Dlk1* depletion altered the myogenic capacity during skeletal muscle regeneration, we performed a detailed qRT-PCR study of myogenic genes in *Dlk1*^{-/-} and *Dlk1*^{+/+} muscles at days 0–14 post-lesion (Fig. 3E). Similar to the *in vitro* situation described above (Fig. 2), we found important myogenic transcription factors, as well as structural muscle proteins,



to be regulated *in vivo* by *Dlk1*, in an inverse pattern to that observed during development (Fig. 1). As such, there were substantially increased amounts of the myogenic transcriptional regulators *Mef2c* and *Meis1*, but also of *Pax7*, in *Dlk1*-depleted animals, whereas no overall difference was seen for *Myod1*, *Myf5*, myogenin or myostatin (Fig. 3E). Likewise, mRNA corresponding to the structural myogenic proteins *Des*, *Myh1* and *Myh7* were all highly increased in *Dlk1*^{-/-} animals compared with *Dlk1*^{+/+} controls,

whereas genes corresponding to *Myh2*, *Tnnt2*, *Ncam1* and *Cd34* were unaffected (Fig. 3E). It has recently been shown that lack of *Dlk1* promotes the self-renewing state of PAX7⁺ myogenic precursor cells (Waddell et al., 2010). We therefore determined whether *Dlk1*-null animals had a larger reservoir of myogenic precursors available for postnatal muscle formation. Indeed, we did find higher basal levels of *Pax3*, *Pax7* and *Myf5* in non-lesioned *Dlk1*^{-/-} adult gastrocnemius muscles (Fig. 3F), indicating increased

Fig. 3. *In vivo* adult muscle regeneration is improved in *Dlk1* null mice. (A) Upper panel: qRT-PCR of regenerating adult mouse gastrocnemius muscle obtained from *Dlk1*^{-/-} or *Dlk1*^{+/+} animals at day 0–14 following a knife-cut lesion confirmed peak expression of *Dlk1* mRNA at regeneration day 7 in *Dlk1*^{+/+} animals. Lower panel: DLK1⁺ cells emerging within the lesion zone were CD31 negative. (B) Immunohistochemical staining of collagen I and CD45 during muscle regeneration in *Dlk1*^{-/-} and *Dlk1*^{+/+} animals showing degenerative fibers and entry of hematopoietic CD45⁺ cells in the lesion zone at day 2. At day 7, inflammation had decreased and, by day 14, muscle fiber integrity was almost completely regained. (C) The relative mRNA levels of genes related to apoptosis/proliferation (*Casp3* and *Pcna*) and hematopoietic cells (*Cd45*, *Thy1*, *Cd14* and *Cd68*) were determined by qRT-PCR. (D) HE and Sirius Red stainings revealed near-normal muscle architecture and equal scarring of both genotypes at day 14 following muscle lesion. (E) Relative qRT-PCR of myogenic genes in regenerating *Dlk1*^{-/-} and *Dlk1*^{+/+} muscles at days 0–14. (F) Relative qRT-PCR of early myogenic transcription factors in normal adult gastrocnemius muscles obtained from *Dlk1*^{+/+} (*n*=4) and *Dlk1*^{-/-} (*n*=4) animals. (G) Muscle function in regenerating *Dlk1*^{+/+} and *Dlk1*^{-/-} animals was tested by rotarod performance. The duration (seconds) that mice stay on the rotating rod was recorded at different time points after muscle lesion. (H) The number of myofibers per area was counted in paraffin-embedded *Dlk1*^{-/-} and *Dlk1*^{+/+} muscle sections at day 14 following lesion. At least 600 fibers were counted for each specimen. For qRT-PCR, raw data in A, C and E were normalized against *Gapdh*, *Pgk1* and *Tbp* (qBase-M: 0.354; CV 0.149). Significance was tested by a two-way ANOVA in A, C and E and by one-way ANOVA in F (**P*<0.05, ***P*<0.01, ****P*<0.0001). Error bars indicate s.d.

satellite cell numbers, which might explain the enhanced myogenic differentiation seen in *Dlk1*-null muscles. Taken together, these results point to a novel, and surprising, biological function of *Dlk1* as an inhibitor of adult skeletal muscle regeneration. To test this result functionally, we examined rotarod performance during the regeneration process, and found that *Dlk1*^{-/-} animals increased their ability to run, whereas *Dlk1*^{+/+} animals performed constantly (Fig. 3G). Furthermore, we quantified the number of myofibers 14 days after regeneration, and found *Dlk1*^{-/-} animals to comprise fewer myofibers per area compared with *Dlk1*^{+/+} controls (Fig. 3H), suggesting hypertrophy of *Dlk1*-depleted muscles. This result clearly contradicts our previous data showing muscle hypertrophy in the Callipyge sheep and in transgenic mice that overexpress membrane-bound DLK1 (Davis et al., 2004). We did not, however, find any centrally located nuclei in neither *Dlk1*^{+/+} or *Dlk1*^{-/-} regenerated fibers (data not shown), another trait of the Callipyge phenotype reflecting immature fibers (Davis et al., 2004). Thus, although *Dlk1*^{-/-} muscles are developmentally smaller, they seem to have a higher regenerative capacity, which might be due to increased numbers of PAX7⁺ progenitors, as suggested by others (Waddell et al., 2010) and supported herein. Notably, we also observed an overall decrease in the vascular gene expression program during the regeneration process (data not shown), again suggesting that *Dlk1* is likely to also provide substantial enhancement of vascularization during *in vivo* muscle remodeling, as also seen *in vitro*.

DLK1-overexpressing myogenic cells display enlarged myofibers *in vitro*

It has recently been shown that overexpressing the membrane-bound DLK1 in myogenic cells promotes myogenic differentiation as visualized by myofiber enlargement and increased myosin expression (Waddell et al., 2010). These results disagree with ours showing enhanced myogenesis of adult *Dlk1*^{-/-} muscle. To clarify

this contradiction, we overexpressed *Dlk1* in myogenic cells. Since both *Dlk1* isoforms are available *in vivo*, we overexpressed full-length *Dlk1* giving rise to both membrane and soluble DLK1, and examined myogenic differentiation. Myogenic cells transfected with empty vector did not show any expression of *Dlk1* at the mRNA or protein level, whereas high amounts of *Dlk1* were seen in cells transfected with *Dlk1* vector (Fig. 4A,B). In agreement with Waddell et al. (Waddell et al., 2010), we found that DLK1-expressing cultures exhibited enlarged myofibers (Fig. 4C) as well as higher levels of myosin (Fig. 4D). However, by culturing myogenic cells with soluble DLK1-conditioned medium, we found the enhanced myogenic phenotype to be caused by the soluble form of DLK1, at least in the case of increased myosin expression (Fig. 4E). These results together suggest that *Dlk1* promotes myogenic differentiation of myogenic precursors, which is in contrast to the data described above for adult *Dlk1*^{-/-} muscle. However, by looking at *Myod1*, which is one of the earliest markers of myogenic commitment, we found that overexpression of *Dlk1* resulted in lower *Myod1* (Fig. 4F), a result that supports the higher *Myod1* level seen within *Dlk*-depleted MDC cultures (Fig. 2E). Yet, no change was found in *Myod1* levels in myogenic cells exposed to DLK1-conditioned medium (Fig. 4G). These data might indicate that the membrane-bound and soluble DLK1 have different effects on the distinct stages of myogenesis, a result that is in agreement with the many distinct effects of DLK1 on adipogenesis. Further complicating this, we found that different levels of DLK1 seem to have opposite effects on *Myod1* levels (Fig. 4H). Thus, as known from studies of *Dlk1* in adipogenesis (Garcés et al., 1999), the exact local level of DLK1 appears to be of great importance for the specific myogenic outcome and this might explain why several contradictions exist within others' and our own data on *Dlk1* in muscle.

Dlk1-null mice display substantial levels of the adipokine adiponectin during skeletal muscle regeneration without concomitant regulation of adipogenic conversion

Whereas the function of DLK1 in muscle is under intense investigation, its role as an important regulator of adipogenic differentiation is well established (Moon et al., 2002). Because adipose deposition can be seen in skeletal muscle disease, probably as a result of pathological differentiation of myogenic-residing precursors into adipocytes (Vettor et al., 2009), we speculated whether *Dlk1* regulates adipogenic commitment of muscle-residing progenitors. Indeed, we did observe a substantial increase in adiponectin, an adipokine known to play an important role in adipogenesis (Fu et al., 2005), during *in vitro* and *in vivo* *Dlk1*^{-/-} muscle formation (Fig. 5A,B). We therefore compared the adipogenic differentiation capacity of MDCs from day 7 post-lesioned *Dlk1*^{-/-} and *Dlk1*^{+/+} m. gastrocnemius. As determined by phase microscopy, lipid-filled cells appeared at day 3 in both *Dlk1*^{-/-} and *Dlk1*^{+/+} MDC cultures, and their number increased progressively over time with adipogenic cells dominating the cultures at day 10 (Fig. 5C). The expression of numerous white and brown adipose tissue genes was highly increased at day 10 versus day 0 cultures (Fig. 5D), which confirmed substantial adipogenic differentiation. Yet, besides moderate differences in *Bmp7*, *Cidea* and *Cebpd*, no significant changes in gene expression were observed between *Dlk1*^{-/-} and *Dlk1*^{+/+} MDCs, and this result was supported by Oil Red O quantification of triglycerides and lipids (Fig. 5E,F). Furthermore, we elucidated whether *Dlk1* affected adipose differentiation and deposition during *in vivo* muscle

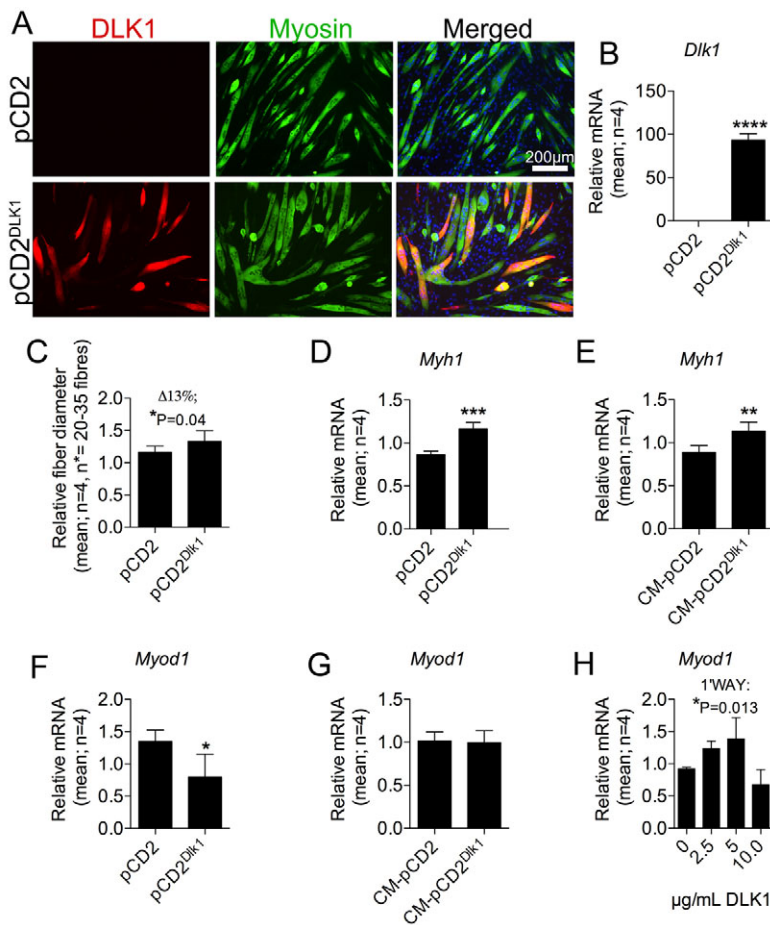


Fig. 4. *Dlk1*-overexpressing myogenic cells reveal larger myofibers. (A) Myogenic cells were transfected with a *Dlk1*-containing (pCD2^{DLK1}) or empty control (pCD2) vector and immediately induced to differentiate. Immunocytochemistry for DLK1 (red) verified the overexpression of DLK1 in pCD2^{DLK1} myogenic cells and the absence of endogenous DLK1 in the pCD2-treated myogenic cells. Myosin (MF20, green) was used to identify myofibers in the cultures. Nuclei were stained blue with DAPI. (B) qRT-PCR verified the overexpression of *Dlk1* in pCD2^{DLK1}-transfected cultures. (C) Fiber diameters were measured in pCD2^{DLK1}- and pCD2-transfected cultures ($n=4$ with 20–35 fibers each). (D,E) qRT-PCR revealed a significant upregulation of *Myh1* mRNA in *Dlk1*-transfected myoblasts (D) and also in myoblasts cultured with conditioned medium from the *Dlk1* transfectants (E). (F,G) mRNA levels of *Myod1* were significantly lower in pCD2^{DLK1} cultures (F) but unchanged when normal myogenic cells were cultured in the presence of soluble DLK1 (G). (H) Myogenic cells were cultured in the presence of varying amounts of DLK1. Significance was tested by a one-way ANOVA (H), a paired *t*-test (C) or Student's *t*-test in (B,D–G) (* $P<0.05$, ** $P<0.01$, *** $P<0.0001$). Error bars indicate s.d.

remodeling. Although *Bmp7* levels were increased in *Dlk1*^{−/−} compared with *Dlk1*^{+/+} animals more or less throughout the regeneration period, no overall changes were observed in *Pparg*, *Cebpd*, *Cidea* and *Ppargc1a* gene expression (Fig. 5G). However, *Prdm16*, a factor recently shown to promote brown adipogenesis over myogenesis from a common (brown adipogenic/myogenic) progenitor (Yin et al., 2013), was lower in *Dlk1*^{−/−} muscles during regeneration compared with *Dlk1*^{+/+} muscles (Fig. 5G). These data thus support the results that *Dlk1*-null mice have an enhanced myogenic capacity. Oil Red O staining (Fig. 5H) of regenerating muscle at days 2, 7 and 14 confirmed that lipid deposition was equal and nearly absent in both genotypes. Thus, although adiponectin levels were highly elevated in regenerating *Dlk1*^{−/−} muscles, *Dlk1* does not seem to be involved in intramuscular adipogenic differentiation. This indicates that the muscle fibers, rather than adipogenic precursors/adipocytes, are the source of adiponectin, as shown by others (Krause et al., 2008). These results thus exclude the possibility that the enhanced myogenic capacity seen in *Dlk1*-null mice is due to an altered adipogenic-myogenic commitment of muscle precursors.

DISCUSSION

We, and others, have shown numerous data consolidating a role for *Dlk1* in developing, regenerating and diseased muscle, yet the function of this gene in muscle biology remains to be elucidated. One recent study (Waddell et al., 2010) in which *Dlk1* was depleted in the myogenic cell lineage (using a *Myf5-Cre* mouse) suggests that *Dlk1* is necessary for and promotes proper muscle regeneration. However, our results imply a far more complex role for *Dlk1* in

skeletal muscle, as a novel myogenic factor that, on the one hand, promotes muscle development but, on the other hand, inhibits muscle regeneration. To our knowledge, such a dual role of a protein has never been reported for skeletal muscle, a tissue that in its adult stage is, so far, known to recapitulate its fetal myogenic program upon damage (Zammit et al., 2006). Indeed, we did find a temporary re-expression of the myogenic gene program during muscle regeneration, but this was substantially enhanced in *Dlk1*-null muscle, suggesting that *Dlk1* might act as a brake on adult muscle regeneration. This result is contrary to some of the results reported by Waddell et al. (Waddell et al., 2010). However, as *Dlk1* is foremost expressed in non-myogenic cells, the study by Waddell et al. only achieved a modest 35% reduction in muscle *Dlk1* using their *Myf5*-dependent *Dlk1* knockdown strategy, whereas we used mice with complete absence of *Dlk1*. It is known that changes in the levels of *Dlk1* might affect the differentiation outcome (Bauer et al., 1998; Nueda et al., 2007) and the relative large amount of muscle-residing *Dlk1* remaining in the Waddell et al. study is thus likely to have a rather large effect. In support of that, we found that low amounts of DLK1 enhanced the levels of the early myogenic transcription factor *Myod1*, whereas higher DLK1 levels had the opposite effect. Furthermore, we used a stab lesion model in m. gastrocnemius whereas Waddell et al. (Waddell et al., 2010) performed cardiotoxin-induced lesion of the tibialis anterior muscle, two parameters (type of muscle and lesion) that are known to have a large impact on the regeneration scheme and might explain some of the differences observed for the role of *Dlk1* in muscle regeneration. Alternatively, different functions of DLK1 protein variants have been described (Ferrón et al., 2011; Garcés et al.,

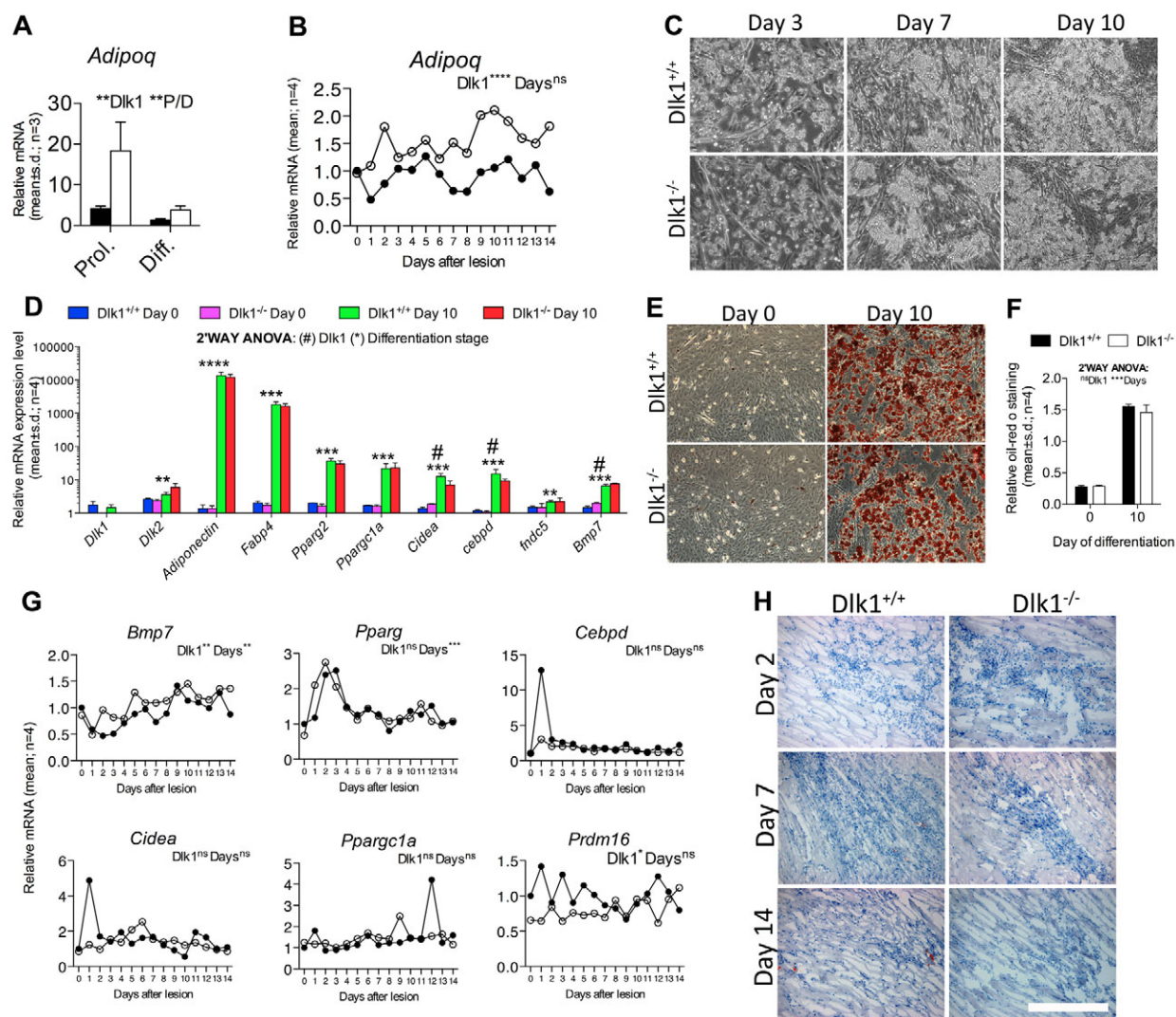


Fig. 5. *Dlk1* exerts no effect on the *in vitro* and *in vivo* adipogenic potential of muscle-residing cells. (A,B) Relative qRT-PCR revealed that the expression of adiponectin was significantly higher in *Dlk1*^{-/-} MDCs [both proliferating (P) and differentiating (D)] compared with *Dlk1*^{+/+} controls (A) as well as in *Dlk1*^{-/-} versus *Dlk1*^{+/+} regenerating adult mouse gastrocnemius muscle (B). (C) Phase microscopy of cultured *Dlk1*^{-/-} and *Dlk1*^{+/+} MDCs during a 10-day adipogenic induction scheme. (D) qRT-PCR of several white and brown adipose tissue-related genes in *Dlk1*^{-/-} and *Dlk1*^{+/+} MDCs at day 0 and 10 of *in vitro* adipogenesis. (E,F) Oil Red O staining (E) and quantification (F) of *Dlk1*^{-/-} and *Dlk1*^{+/+} MDCs at confluence (day 0) and after 10 days of adipogenic induction. (G) qRT-PCR of white and brown adipose tissue-related genes during *in vivo* regeneration in *Dlk1*^{-/-} and *Dlk1*^{+/+} animals. (H) Oil Red O staining of muscle at days 2, 7 and 14 of regeneration. In agreement with the expression profiles shown in G, deposition of lipids did not differ between genotypes. Error bars indicate s.d. Statistical significance was tested by a two-way ANOVA (**P*<0.05, ***P*<0.01, ****P*<0.0001, *****P*<0.00001). For qRT-PCR, raw data were normalized against (A) *18s* and *Tbp* (qBase-M: 0.258; CV 0.089), (D) *Gapdh*, *Actb* and *Rpl13a* (qBase-M: 1.453; CV 0.467), and (G) *Gapdh*, *Pgk1* and *Tbp* (qBase-M: 0.354; CV 0.149). Scale bar: 500 μ m.

1999) and the proper equilibrium of expression among them might affect the outcome. In this sense, it is unclear how the knockdown strategy used by Waddell et al. affected this equilibrium. In relation to this, we found that only the soluble form of DLK1 was sufficient for myosin enhancement, whereas *Myod1* levels depended on the exact level of DLK1. This clearly points to a complex system in which fine-tuning of DLK1 may have a large impact on the myogenic outcome, and which might explain the contradictions regarding *Dlk1* in muscle that have been reported herein and by others (Waddell et al., 2010). Finally, *Dlk1* is expressed by both the satellite/muscle cell itself and interstitial cells depending on the developmental stage, and secreted protein variants are also expressed by other more distant cells. Thus, DLK1 might act upon

the muscle cell by direct cell-cell contact or indirectly by a paracrine or exocrine mechanism. In agreement with that hypothesis, our data show that during development *Dlk1* is expressed in the muscle cells where it seems to promote myogenesis, a scenario that is similar to that observed by Waddell et al. This is further supported by previous studies showing that overexpression of the ovine *Dlk1* in mouse muscle cells dramatically stimulates myogenesis (Davis et al., 2004). Collectively, the combined data indicate dual functions for cell-autonomous and non-cell-autonomous DLK1 activity in skeletal muscle, a scenario also demonstrated for NOTCH1 (Becam et al., 2010). In skeletal muscle, NOTCH1 activation is necessary for the specification of myogenic stem cells and maintenance of the transit-amplifying state whereas the decay of NOTCH signaling is

a prerequisite for fusion and terminal differentiation of myoblasts (Gildor et al., 2012; Mourikis et al., 2012). As such, constitutive NOTCH1 activation in *Myf5*-derived cells leads to a complete absence of skeletal muscle in *Myf5^{Cre/+}:R26R^{stop-NICD-nGFP/+}* embryos and a dramatic increase in *Pax7⁺/Myf5⁺/Myog⁻* myogenic stem/progenitor cells (Mourikis et al., 2012). Strikingly in this regard, we observed that *Dlk1*^{-/-} animals express higher levels of *Pax3*, *Pax7* and *Myf5* both during development and in adulthood, indicative of an enhanced stem cell pool in these animals. This result is in agreement with Waddell et al. who demonstrated that the lack of *Dlk1* promotes the number of *Pax7⁺* myogenic precursors (Waddell et al., 2010). Provided that these stem cells have a normal postnatal differentiation potential, this may explain why *Dlk1*^{-/-} mice reveal a much lower muscle mass but improved regenerative abilities. Whether this is a consequence of alterations in NOTCH signaling owing to the complete lack of DLK1 (as one of several NOTCH1 antagonists) is not clear. Although the *Dlk1*^{-/-} skeletal muscle phenotype to some extent could mirror the activation of NOTCH1, we did not observe any conclusive differences in the downstream effectors HES1 and HEY1 to support this hypothesis (data not shown). Although DLK1 in skeletal muscle might act independently of NOTCH1, as recently suggested for neurogenesis (Ferrón et al., 2011), we cannot exclude a role for NOTCH1 herein based on the present data.

In summary, the identification of DLK1 as a novel important myogenic factor that both promotes skeletal muscle development while inhibiting adult skeletal muscle remodeling may open up new possibilities of targeting muscle disease in human patients. The induced greater myogenic commitment of muscle precursors lacking *Dlk1* might be mirrored by shutting down DLK1 either by antibodies or gene therapy, or by directing a compound against DLK1 or the still controversial interaction partner of DLK1. Moreover, we believe that the paradoxical way in which DLK1 regulates myogenesis should caution scientists not to define a protein function merely by its ability to affect the phenotype of the embryo, which in the case of muscle-residing *Dlk1* is clearly inaccurate. Thus, although regeneration is often a recapitulation of embryogenesis, it may also be the reverse with many unrecognized functions of developmental proteins.

Acknowledgements

We would like to thank Charlotte Nielsen, Tonja L. Jørgensen and Anette Kliem (LMCC, Odense University Hospital) for technical assistance on this study; associate professor Kate Lykke Lambertsen for use and help with Rotarod experiments; and Gunnhildur Asta Traustadottir for help with plasmid handling.

Funding

This work was supported by the Novo Nordisk Foundation; The Danish National Research Council; the Aase and Ejnar Danielsens Foundation; The Lundbeck Foundation; the Hertha Christensens Foundation; the Eva and Henry Frønkels Foundation; the A. P. Møller Foundation for the Advancement of Medical Science; and the Department of Clinical Biochemistry and Pharmacology at Odense University Hospital.

Competing interests statement

The authors declare no competing financial interests.

Author contributions

C.H.J. and D.C.A. conceived and designed the experiments; collected, analyzed and interpreted the data; and prepared the manuscript. V.B. collected data, M.K. and J.L. interpreted and discussed the data; and edited the manuscript. S.P.S. interpreted the data.

Supplementary material

Supplementary material available online at <http://dev.biologists.org/lookup/suppl/doi:10.1242/dev.095810/-/DC1>

References

- Abdallah, B. M., Ditzel, N., Mahmood, A., Isa, A., Traustadottir, G. A., Schilling, A. F., Ruiz-Hidalgo, M. J., Laborda, J., Ameling, M. and Kassem, M. (2011). DLK1 is a novel regulator of bone mass that mediates estrogen deficiency-induced bone loss in mice. *J. Bone Miner. Res.* **26**, 1457-1471.
- Andersen, D. C., Schröder, H. D. and Jensen, C. H. (2008). Non-cultured adipose-derived CD45- side population cells are enriched for progenitors that give rise to myofibres in vivo. *Exp. Cell Res.* **314**, 2951-2964.
- Andersen, D. C., Petersson, S. J., Jørgensen, L. H., Bollen, P., Jensen, P. B., Teisner, B., Schroeder, H. D. and Jensen, C. H. (2009). Characterization of DLK1+ cells emerging during skeletal muscle remodeling in response to myositis, myopathies, and acute injury. *Stem Cells* **27**, 898-908.
- Andersen, D. C., Nielsen, C., Jensen, C. H. and Sheikh, S. P. (2013). Horse serum reduces expression of membrane-bound and soluble isoforms of the preadipocyte marker Delta-like 1 homolog (Dlk1), but is inefficient for adipogenic differentiation of mouse preadipocytes. *Acta Histochem.* **115**, 401-406.
- Bachmann, E., Krogh, T. N., Højrup, P., Skjødt, K. and Teisner, B. (1996). Mouse fetal antigen 1 (mFA1), the circulating gene product of mdk, pref-1 and SCP-1: isolation, characterization and biology. *J. Reprod. Fertil.* **107**, 279-285.
- Baladrón, V., Ruiz-Hidalgo, M. J., Bonvini, E., Gubina, E., Notario, V. and Laborda, J. (2002). The EGF-like homeotic protein dlk affects cell growth and interacts with growth-modulating molecules in the yeast two-hybrid system. *Biochem. Biophys. Res. Commun.* **291**, 193-204.
- Bauer, S. R., Ruiz-Hidalgo, M. J., Rudikoff, E. K., Goldstein, J. and Laborda, J. (1998). Modulated expression of the epidermal growth factor-like homeotic protein dlk influences stromal-cell-pre-B-cell interactions, stromal cell adipogenesis, and pre-B-cell interleukin-7 requirements. *Mol. Cell. Biol.* **18**, 5247-5255.
- Becam, I., Fiuza, U. M., Arias, A. M. and Milán, M. (2010). A role of receptor Notch in ligand cis-inhibition in Drosophila. *Curr. Biol.* **20**, 554-560.
- Black, B. L. and Olson, E. N. (1998). Transcriptional control of muscle development by myocyte enhancer factor-2 (MEF2) proteins. *Annu. Rev. Cell Dev. Biol.* **14**, 167-196.
- Chen, L., Qanie, D., Jafari, A., Taipaleenmaki, H., Jensen, C. H., Säämänen, A. M., Sanz, M. L., Laborda, J., Abdallah, B. M. and Kassem, M. (2011). Delta-like 1/fetal antigen-1 (Dlk1/FA1) is a novel regulator of chondrogenic cell differentiation via inhibition of the Akt kinase-dependent pathway. *J. Biol. Chem.* **286**, 32140-32149.
- Davis, E., Jensen, C. H., Schroder, H. D., Farnir, F., Shay-Hadfield, T., Kliem, A., Cockett, N., Georges, M. and Charlier, C. (2004). Ectopic expression of DLK1 protein in skeletal muscle of padumal heterozygotes causes the callipyge phenotype. *Curr. Biol.* **14**, 1858-1862.
- Evans, W. J. (2010). Skeletal muscle loss: cachexia, sarcopenia, and inactivity. *Am. J. Clin. Nutr.* **91**, 1123S-1127S.
- Ferrón, S. R., Charalambous, M., Radford, E., McEwen, K., Wildner, H., Hind, E., Morante-Redolat, J. M., Laborda, J., Guillemot, F., Bauer, S. R. et al. (2011). Postnatal loss of Dlk1 imprinting in stem cells and niche astrocytes regulates neurogenesis. *Nature* **475**, 381-385.
- Floridon, C., Jensen, C. H., Thorsen, P., Nielsen, O., Sunde, L., Westergaard, J. G., Thomsen, S. G. and Teisner, B. (2000). Does fetal antigen 1 (FA1) identify cells with regenerative, endocrine and neuroendocrine potentials? A study of FA1 in embryonic, fetal, and placental tissue and in maternal circulation. *Differentiation* **66**, 49-59.
- Fu, Y., Luo, N., Klein, R. L. and Garvey, W. T. (2005). Adiponectin promotes adipocyte differentiation, insulin sensitivity, and lipid accumulation. *J. Lipid Res.* **46**, 1369-1379.
- Garcés, C., Ruiz-Hidalgo, M. J., Bonvini, E., Goldstein, J. and Laborda, J. (1999). Adipocyte differentiation is modulated by secreted delta-like (dlk) variants and requires the expression of membrane-associated dlk. *Differentiation* **64**, 103-114.
- Gildor, B., Schejter, E. D. and Shilo, B. Z. (2012). Bidirectional Notch activation represses fusion competence in swarming adult Drosophila myoblasts. *Development* **139**, 4040-4050.
- Hellemans, J., Mortier, G., De Paepe, A., Speleman, F. and Vandesompele, J. (2007). qBase relative quantification framework and software for management and automated analysis of real-time quantitative PCR data. *Genome Biol.* **8**, R19.
- Jensen, C. H., Krogh, T. N., Højrup, P., Clausen, P. P., Skjødt, K., Larsson, L. I., Enghild, J. J. and Teisner, B. (1994). Protein structure of fetal antigen 1 (FA1). A novel circulating human epidermal-growth-factor-like protein expressed in neuroendocrine tumors and its relation to the gene products of dlk and pG2. *Eur. J. Biochem.* **225**, 83-92.
- Krause, M. P., Liu, Y., Vu, V., Chan, L., Xu, A., Riddell, M. C., Sweeney, G. and Hawke, T. J. (2008). Adiponectin is expressed by skeletal muscle fibers and influences muscle phenotype and function. *Am. J. Physiol.* **295**, C203-C212.
- Laborda, J., Sausville, E. A., Hoffman, T. and Notario, V. (1993). dlk, a putative mammalian homeotic gene differentially expressed in small cell lung carcinoma and neuroendocrine tumor cell line. *J. Biol. Chem.* **268**, 3817-3820.

- McFadyen, M. P., Kusek, G., Bolivar, V. J. and Flaherty, L. (2003). Differences among eight inbred strains of mice in motor ability and motor learning on a rotorod. *Genes Brain Behav.* **2**, 214-219.
- Mei, B., Zhao, L., Chen, L. and Sul, H. S. (2002). Only the large soluble form of preadipocyte factor-1 (Pref-1), but not the small soluble and membrane forms, inhibits adipocyte differentiation: role of alternative splicing. *Biochem. J.* **364**, 137-144.
- Moon, Y. S., Smas, C. M., Lee, K., Villena, J. A., Kim, K. H., Yun, E. J. and Sul, H. S. (2002). Mice lacking paternally expressed Pref-1/Dlk1 display growth retardation and accelerated adiposity. *Mol. Cell. Biol.* **22**, 5585-5592.
- Mourikis, P., Gopalakrishnan, S., Sambasivan, R. and Tajbakhsh, S. (2012). Cell-autonomous Notch activity maintains the temporal specification potential of skeletal muscle stem cells. *Development* **139**, 4536-4548.
- Nueda, M. L., Baladrón, V., Sánchez-Solana, B., Ballesteros, M. A. and Laborda, J. (2007). The EGF-like protein dlk1 inhibits notch signaling and potentiates adipogenesis of mesenchymal cells. *J. Mol. Biol.* **367**, 1281-1293.
- Raghunandan, R., Ruiz-Hidalgo, M., Jia, Y., Ettinger, R., Rudikoff, E., Riggins, P., Farnsworth, R., Tesfaye, A., Laborda, J. and Bauer, S. R. (2008). Dlk1 influences differentiation and function of B lymphocytes. *Stem Cells Dev.* **17**, 495-507.
- Rodríguez, P., Higuera, M. A., González-Rajal, A., Alfranca, A., Fierro-Fernández, M., García-Fernández, R. A., Ruiz-Hidalgo, M. J., Monsalve, M., Rodríguez-Pascual, F., Redondo, J. M. et al. (2012). The non-canonical NOTCH ligand DLK1 exhibits a novel vascular role as a strong inhibitor of angiogenesis. *Cardiovasc. Res.* **93**, 232-241.
- Saini, A. and Stewart, C. E. (2006). Adult stem cells: the therapeutic potential of skeletal muscle. *Curr. Stem Cell Res. Ther.* **1**, 157-171.
- Smas, C. M., Green, D. and Sul, H. S. (1994). Structural characterization and alternate splicing of the gene encoding the preadipocyte EGF-like protein pref-1. *Biochemistry* **33**, 9257-9265.
- Vandesompele, J., De Preter, K., Pattyn, F., Poppe, B., Van Roy, N., De Paepe, A. and Speleman, F. (2002). Accurate normalization of real-time quantitative RT-PCR data by geometric averaging of multiple internal control genes. *Genome Biol.* **3**, RESEARCH0034.
- Vettor, R., Milan, G., Franzin, C., Sanna, M., De Coppi, P., Rizzuto, R. and Federspil, G. (2009). The origin of intermuscular adipose tissue and its pathophysiological implications. *Am. J. Physiol.* **297**, E987-E998.
- Waddell, J. N., Zhang, P., Wen, Y., Gupta, S. K., Yevtdiyenko, A., Schmidt, J. V., Bidwell, C. A., Kumar, A. and Kuang, S. (2010). Dlk1 is necessary for proper skeletal muscle development and regeneration. *PLoS ONE* **5**, e15055.
- Yan, Z., Choi, S., Liu, X., Zhang, M., Schageman, J. J., Lee, S. Y., Hart, R., Lin, L., Thurmond, F. A. and Williams, R. S. (2003). Highly coordinated gene regulation in mouse skeletal muscle regeneration. *J. Biol. Chem.* **278**, 8826-8836.
- Yevtdiyenko, A. and Schmidt, J. V. (2006). Dlk1 expression marks developing endothelium and sites of branching morphogenesis in the mouse embryo and placenta. *Dev. Dyn.* **235**, 1115-1123.
- Yin, H., Pasut, A., Soleimani, V. D., Bentzinger, C. F., Antoun, G., Thorn, S., Seale, P., Fernando, P., van Ijcken, W., Grosveld, F. et al. (2013). MicroRNA-133 controls brown adipose determination in skeletal muscle satellite cells by targeting Prdm16. *Cell Metab.* **17**, 210-224.
- Zammit, P. S., Partridge, T. A. and Yablonka-Reuveni, Z. (2006). The skeletal muscle satellite cell: the stem cell that came in from the cold. *J. Histochem. Cytochem.* **54**, 1177-1191.